$$(\mathbf{A} \fatsemi \mathbf{B}) \otimes (\mathbf{C} \fatsemi \mathbf{D}) = (\mathbf{BA}) \otimes (\mathbf{DC}) \tag{1}$$

$$= \begin{bmatrix} \mathbf{BA} & \mathbf{0} \\ \mathbf{0} & \mathbf{DC} \end{bmatrix} \tag{2}$$